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1. Real numbers $a_1, a_2, \dots, a_n$ are given. For each i ($1 \leq i \leq n$) define

$$d_i = \max\{a_j : 1 \leq j \leq i\} - \min\{a_j : i \leq j \leq n\}$$

and let

$$d = \max\{d_i : 1 \leq i \leq n\}.$$

- (a) Prove that for any real numbers $x_1 \leq x_2 \leq \dots \leq x_n$,

$$\max\{|x_i - a_i| : 1 \leq i \leq n\} \geq \frac{d}{2}.$$

- (b) Show that there exist real numbers $x_1 \leq x_2 \leq \dots \leq x_n$ such that equality holds.

2. Consider five points A, B, C, D, E such that $ABCD$ is a parallelogram and $BCED$ is a cyclic quadrilateral. Let ℓ be a line passing through A . Suppose that ℓ intersects segment DC at F and line BC at G . Suppose also that $EF = EG = EC$. Prove that ℓ bisects angle DAB .
3. In a mathematical competition some competitors are friends. Friendship is mutual. A group of competitors is called a clique if each two of them are friends. Given that the largest size of a clique is even, prove that the competitors can be arranged in two rooms so that the largest clique size in each room is the same.
4. In triangle ABC the bisector of angle BCA meets the circumcircle again at R , the perpendicular bisector of BC at P , and that of AC at Q . Let K and L be the midpoints of BC and AC respectively. Prove that triangles RPK and RQL have equal area.

5. Let a and b be positive integers. Show that if $4ab - 1$ divides $(4a^2 - 1)^2$, then $a = b$.
6. Let n be a positive integer. Let

$$S = \{(x, y, z) : x, y, z \in \{0, 1, \dots, n\}, x + y + z > 0\}.$$

Determine the minimum number of planes whose union contains S but does not contain $(0, 0, 0)$.